

Case:

Exploratory Data Analysis in Online Casinos

Revenue and Performance

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Analyze online casino performance metrics

Connecticut Online Casino Consortium

Data Source: monthly license revenue reports

Methodology

1. Processing: Cleaning, date conversion, create metrics
2. Exploratory analysis: view revenue, licences, promotion.
3. Descriptive analysis: correlation and insights.
4. Business Recommendation



The Problems



1. Revenue Trends Analysis:
 - a. How has the total gaming revenue changed over time?
 - b. Are there seasonal trends in casino revenue?
2. Impact of Promotions:
 - a. Do promotional credits and deductions lead to higher overall revenue?
 - b. Is there a correlation between promotional spending and net winnings?
3. Casino Performance Comparison:
 - a. Which casino licenses generate the highest revenue?
 - b. Are there specific casinos consistently underperforming?
4. Wagering and Payout Patterns:
 - a. How do cancelled wagers impact revenue?
 - b. What percentage of wagers is typically returned as winnings?

DATA SET DESCRIPTION

The dataset contains 143 records with 11 columns.

There are no null values, so imputation is not required.

The data ranges from October 2021 to 2024.

Data columns (total 11 columns):			
#	Column	Non-Null Count	Dtype
0	Fiscal Year	143 non-null	object
1	Month Ending	143 non-null	object
2	Licensee	143 non-null	object
3	Wagers	143 non-null	int64
4	Patron Winnings	143 non-null	int64
5	Cancelled Wagers	143 non-null	int64
6	Online Casino Gaming Win/(Loss)	143 non-null	int64
7	Promotional Coupons or Credits Wagered (3)	143 non-null	int64
8	Promotional Deduction (4)	143 non-null	int64
9	Total Gross Gaming Revenue	143 non-null	int64
10	Payment (5)	143 non-null	int64

	Wagers	Patron Winnings	Cancelled Wagers	Online Casino Gaming Win/(Loss)
count	1.430000e+02	1.430000e+02	1.430000e+02	1.430000e+02
mean	2.742207e+08	2.655218e+08	1.783408e+05	8.520509e+06
std	2.815530e+08	2.725166e+08	2.826004e+05	8.921912e+06
min	1.145280e+05	1.113650e+05	0.000000e+00	-8.099000e+04
25%	1.407190e+06	1.357855e+06	5.000000e+00	3.713950e+04
50%	2.656540e+08	2.575127e+08	3.887000e+03	7.866081e+06
75%	5.352846e+08	5.156584e+08	2.913700e+05	1.587461e+07
max	7.838842e+08	7.573176e+08	1.133136e+06	2.570443e+07



FASE: EDA

Exploratory Data Analysis

Q1. Revenue Trends Analysis

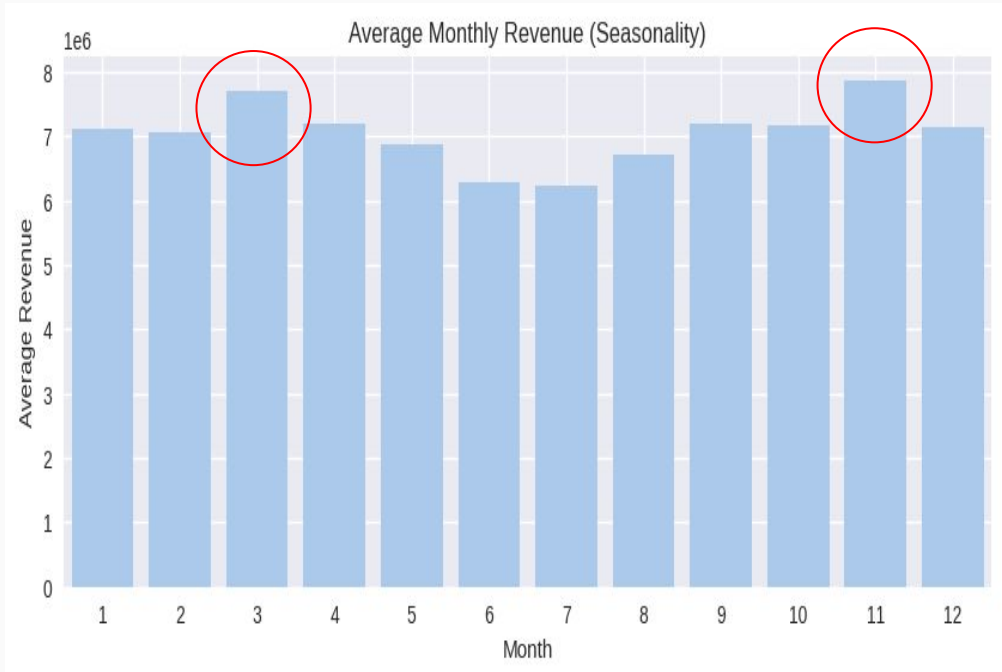
1. How has the total gaming revenue changed over time?



2. Are there seasonal trends in casino revenue?

- ★ The total revenue has shown increasing patterns from October 2021 onwards, an stability from Jan 2022 to Sep 2022, then increasing fluctuation almost of 200% of growing.
- ★ Peaks are observed in **March, November**, suggesting potential **seasonality** (holidays or promotions). People have more free time and are more willing to spend on entertainment such as online gaming.
- ★ Overall trend shows an overall **upward trajectory**, showing long-term market growth

Seasonality Analysis (monthly)



There is a pattern indicating slightly higher revenues in months like **March and November**, suggesting a potential seasonal behavior.

Q2: Impact of Promotions

1. Do promotional credits and deductions lead to higher overall revenue?



Yes, there is evidence that **promotional credits and deductions are positively associated with higher gross gaming revenue**. Based on the correlation analysis:

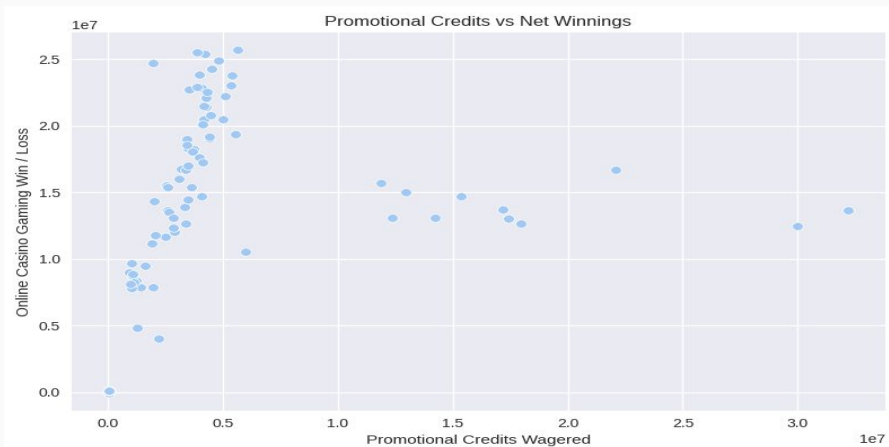
- ★ **Promotional Deductions** have a **very strong positive correlation (0.94)** with **Total Gross Gaming Revenue**.
- ★ **Promotional Credits Wagered** have a **moderate positive correlation (0.44)** with gross revenue.

This suggests that when casinos **invest more in promotions**, they generally see **higher revenues**, likely because such promotions encourage more play and attract more users.

	Total Gross Gaming Revenue
Promotional Coupons or Credits Wagered (3)	0.44
Promotional Deduction (4)	0.94
Total Gross Gaming Revenue	1.00

Q2: Impact of Promotions

2. Is there a correlation between promotional spending and net winnings?



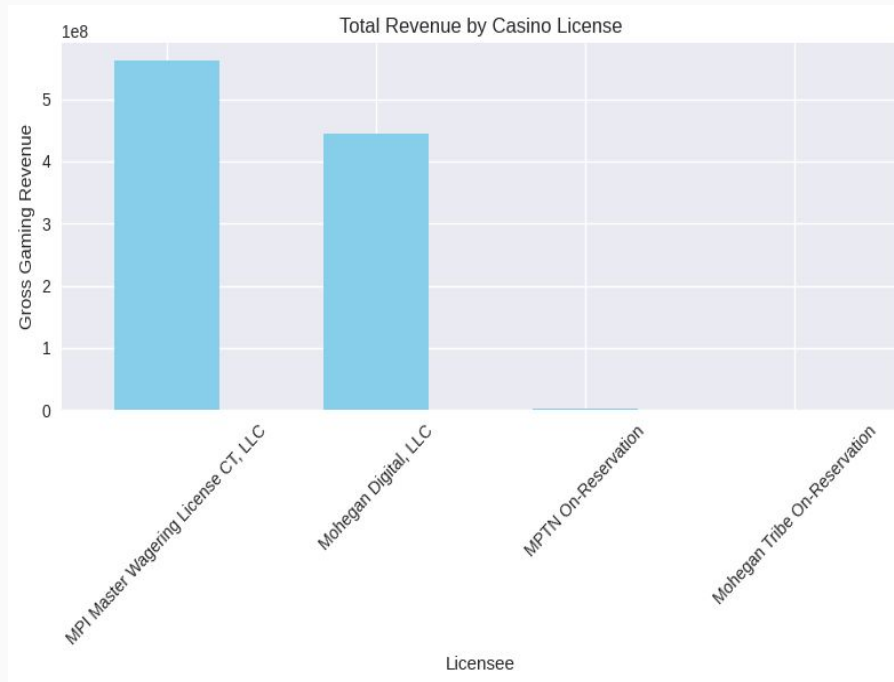
- ★ There is a moderate **positive correlation (0.47)** between promotional credits wagered and net winnings.
- ★ This suggests that when players **use promo credits**, they tend to generate higher net profits for the casino.
- ★ Promotions not only attract more wagers but may also influence player behavior in ways that increase casino profitability.

Correlation between promotional credits and net winnings:

Promotional Coupons or Credits Wagered (3)	1.00
Online Casino Gaming Win/(Loss)	0.47
Promotional Coupons or Credits Wagered (3)	0.47
Online Casino Gaming Win/(Loss)	1.00

Q3: Casino Performance Comparison

5. Which casino licenses generate the highest revenue?



- ★ MPI Master Wagering License CT, LLC is the top-performing casino license with over \$561 million in gross gaming revenue.
- ★ Mohegan Digital, LLC follows closely with \$444 million.
- ★ The two on-reservation licenses (MPTN and Mohegan Tribe) have significantly lower revenue.

Q3: Casino Performance Comparison

6. Are there specific casinos consistently underperforming??



Yes. Based on the **monthly revenue trends** shown in the line chart:

- ★ **MPTN On-Reservation** and **Mohegan Tribe On-Reservation** show consistent underperformance over time.
- ★ Their revenue is significantly lower than **MPI** and **Mohegan Digital**, with no notable spikes or upward trend.

Q4: Wagering & Payout Patterns

7. How do cancelled wagers impact revenue?



Yes. **strong positive correlation of 0.83** between **Cancelled Wagers** and **Gross Gaming Revenue**:

- ★ Cancelled wagers show a **strong positive correlation** with gross revenue.
- ★ This means that **higher total revenue months** tend to also have **more cancelled wagers**.
- ★ However, this **does not mean cancellations increase revenue**, but rather that **they are more common when activity is higher overall**.
- ★ In other words, **cancelled wagers are not harmful**, but a **byproduct of increased wagering activity**.

Cancelled Wagers Total Gross Gaming Revenue

Cancelled Wagers	1.00	0.83
Total Gross Gaming Revenue	0.83	1.00

Q4: Wagering & Payout Patterns

8. What percentage of wagers is typically returned as winnings?

Payout Ratio	
count	143.00
mean	0.97
std	0.02
min	0.89
25%	0.96
50%	0.97
75%	0.97
max	1.04

To respond this, i'm using data statistic:

- ★ The average **payout ratio** is **97.0%**, meaning that for every \$100 wagered, approximately **\$97 is returned to players** as winnings.
- ★ The **lowest payout** observed was **89%**, while the **highest** reached **104%**, indicating that in rare cases, players won more than they wagered.
- ★ The **standard deviation** is **2%**, showing **low variability** and suggesting that payout behavior is **very consistent** across months and licensees.

Extra: Business Recommendations

1. Invest Strategically in Promotions

- Promotions are strongly linked to revenue growth.
- Focus on optimizing promo campaigns by targeting peak gaming periods.

2. Monitor Revenue and Wagering Trends Monthly

- Continue tracking performance using visual dashboards and correlation tracking.
- Use data to detect early signals of decline or growth in specific licenses.

3. Maintain High Payout Standards

- Keep the payout ratio near 95%–97% to maintain **player trust and retention**.
- Investigate any months with payout ratios above 100% to ensure sustainability.